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Aaditya Tamrakar

Software Engineer · Quant Developer

Software engineer in Go, Python and TypeScript across trading systems and data platforms. Built 80+ order execution and market making algorithms at Algo Maker, Go microservices with Snowflake and Airflow pipelines at Viasat, and NSE options analytics used by 50+ developers. M.S. Computer Science, UMass Boston.

Work Experience

Viasat Inc. *Software Engineer II* Remote, Austin, US · May 2023 – Dec 2025

- Improved in-flight connectivity with the core platform team; developed a Golang microservice for a server deployed in airplanes for monitoring.
- Added new API endpoints, changes to the Snowflake database, and updated Airflow DAGs.
- Modified the CloudFormation stack to create new infrastructure for data monitoring.

Viasat Inc. *Software Engineer Intern* Remote, Boston, US · May – Aug 2022

- Consolidated functions into a common shared library used across teams; achieved 100% unit test coverage.
- Modified the cloud stack to create new infrastructure for data monitoring; joined full time after graduation.

Algo Maker LLP *Full Stack Developer* Mumbai, IN · Dec 2020 – Aug 2021

- Created and implemented 80+ efficient algorithms to execute trade orders, from time/price based to advanced market making using WebSockets, so it could scale to handle millions in funds.
- Designed a dashboard using Angular and Express.js to visualise data stored in MongoDB, which reduced the time to evaluate risk exposure and execute new orders by 50%.
- Set up the complete platform in AWS, closest to the broker's servers, resulting in 20% faster order execution.

Accio.ai *Software Engineer Intern* Mumbai, IN · Apr 2018 – Sep 2018

- Developed and implemented a data pipeline to process 2 million events per day.
- Designed a REST API to query data for ad payment and reporting.
- Created a searchable schema in MongoDB on the raw analytics data (clicks, impressions, views).
- Developed an automation script using the AWS SDK which cut the time to upload each design from minutes to seconds.

Freelance *Full Stack Developer* 2016 – 2019

- Shipped PHP/Laravel backends, Node.js APIs and Ionic/AngularJS apps for small businesses: food delivery, fleet management, dealership reporting, accounting.

Education

Master of Science, Computer Science CGPA 3.7/4 · 2021 – 2023

University of Massachusetts, Boston

- Graduate Teaching Fellow, CS240 Programming in C, for two semesters to classes of 51 and 67 students.

Projects

Option Chain Analysis · Node.js, REST API 100+ stars on GitHub · Nov 2019

Fetches the live NSE Nifty option chain and classifies every strike by price and open interest change; used by 50+ developers in their own projects.

UMB Trading App · HTML5, JavaScript, Figma, User Interface Design · project site Dec 2021

CS615 project: a simple interface where a beginner can trade options easily.

ML Captcha Solver · scikit-learn, PHP Jan 2018

Image pipeline and classifier that reads Amazon-style captchas, 99.7% reported accuracy on that style.

Open Source

Broker SDKs forked and maintained for my own tooling: Kite Connect (Go), XTS v2 and 5paisa (Node.js), Hummingbot client (Python), option pricing formulas. Open sourced the option chain analyser, futures spread chart and Dream11 team builder.

Skills

Languages	Go, Python, TypeScript, JavaScript, C++, C, PHP, SQL, Java
Trading	Order execution, market making, WebSockets, NSE option chain data, option pricing, Kite Connect, XTS, 5paisa, Hummingbot
Data & infra	AWS, CloudFormation, Snowflake, Apache Airflow, Kafka, Spark, Docker, Kubernetes, Redis, MongoDB, Linux, Playwright